

Data Preparation Pipeline for Distribution-Network Learning

Implemented pipeline · MATPOWER case33bw primary benchmark · case69 external transfer benchmark · processing stops at model training

01 · Benchmark Network Sources

Primary benchmark · case33bw

12.66 kV · 33 buses · 37 branch records
5 normally-open tie branches enable radial reconfiguration

External transfer benchmark · case69

12.66 kV · 69 buses
Structurally distinct radial feeder;
reserved for transfer analysis

Raw electrical fields

Bus demand P_i^0, Q_i^0 ; branch r_{ij}, x_{ij}
base kV/MVA; branch status and
admissible radial topology

02 · Operating-State and Topology Synthesis

Operating-state factors

hour · day-of-year · weekday
global loading $g \sim U(0.72, 1.30)$ + nodal
lognormal variability

Load-state synthesis

$P_i = P_i^0 s_i, \quad Q_i = Q_i^0 s_i \epsilon_i$
daily + seasonal + weekend modulation
stress factor clipped to $s_i \in [0.35, 1.65]$

Topology and source-voltage sampling

case33bw: 60 enumerated radial topologies
 $V_s \sim U(0.975, 1.045)$ p.u.
case69: fixed radial topology; V_s
 $\sim U(0.98, 1.04)$ p.u.

03 · Exact Physics, Targets and Telemetry

Exact electrical state

Radial AC power-flow solver $f_{PF}(P, Q, \mathcal{G}, V_s)$
returns $V_{\min}, V_{\max}, L_{kW}$ and convergence

Scientific targets

$y = [V_{\min}, L_{kW}, c]$
 $c = \mathbf{1}[\text{conv} \wedge 0.95 \leq V_{\min} \wedge V_{\max} \leq 1.05]$

Communication / telemetry degradation

scenario missingness parameter $\rho \sim U(0, 0.20)$
 $o_i \sim \text{Bernoulli}(1 - \rho)$; slack node always observed
age $a_i \in [0, 4]$; noise $\sigma_i = 0.005 + 0.012a_i$;
missing $P_i, Q_i \rightarrow \text{NaN}$

06 · Model-Ready Tensors and Training Entry

Model-ready batch

$X_B \in \mathbb{R}^{B \times 69 \times 10}$
 $A_B \in \{0, 1\}^{B \times 69 \times 69}$
 $m_B \in \{0, 1\}^{B \times 69}$
 $\tilde{z}_B \in \mathbb{R}^{B \times 21}, y_B \in \mathbb{R}^{B \times 3}$

Common comparison protocol

same topology-disjoint roles and targets
engineering baselines · tree/MLP
baselines · graph comparators · ACG-
MAC

MODEL TRAINING STARTS HERE

Only training states update model
parameters

selection chooses configuration;
calibration estimates uncertainty

inspected topology holdout is
evaluated only after model fixation

Prepared data → model training

05 · Topology-Disjoint Roles and Scaling

Primary case33bw roles

Train: 33 topologies / 2,146 states
Selection: 9 / 638 · Calibration: 9 / 598
Inspected topology holdout (test): 9 / 618

External role

case69: 800 states
No primary-test reuse; local adaptation is
evaluated separately

Node-feature pre-scaling

$P/P_{\max}^{ref}, Q/Q_{\max}^{ref}$; degree / max-degree
depth / max-depth; clipped age $a_i/4$

Train-only system normalization

$\tilde{z}_j = (z_j - \mu_{j,tr}) / \sigma_{j,tr}$
 $\sigma_{j,tr} < 10^{-6} \Rightarrow \sigma_{j,tr} = 1$

Target interface

loss uses $\log(1 + L)$ during regression
learning
selection / calibration / test statistics never fit
preprocessing

04 · Imputation and Model Representations

Leakage-safe observation completion

Observed values are retained; missing values
use reference-load shape
 $x_i^{imp} = x_i^{ref} s_{obs}$ only for missing nodes

Node tensor · 10 features per padded node

$X \in \mathbb{R}^{69 \times 10}$
imputed P,Q · nominal P,Q · degree · depth
availability · normalized age · slack flag ·
active-node flag

Graph tensors

$A \in \{0, 1\}^{69 \times 69}, \quad m_n \in \{0, 1\}^{69}$
topology adjacency + real-node mask

System context · 21 features

7 load statistics + 4 operating / telemetry
2 Linear-DistFlow proxies + 8 topology
descriptors
 $z \in \mathbb{R}^{21}$

Scenario package

$\mathcal{S} = \{X, A, m_n, z, y\}$, with $y \in \mathbb{R}^3$